

D2: Record Duration and Welch Segment Length

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How long must we record, and which L should we use?

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SHARC buoy project Code: `runD2RecordDuration.m`, commit 3668736 Results: `Results/D2/D2_20261002_114914`

Summary

Question. For each Welch segment length L , how does record duration trade random error against finite-resolution (Welch) bias, and what is the shortest record that meets the pre-registered Identifiable criteria? **Outcome (LSM6DSV16X).** The shortest Identifiable record is **17.1 min with $L = 2048$** ; $L = 2048$ stays Identifiable from 17.1 to 281.5 min, so the 109 min production record sits well inside its window (6.4 times the minimum). $L = 1024$ reaches Identifiable equally early but only until 29 min; $L = 512$ is never Identifiable for any noise case. **Two regimes.** Noisy sensors need longer records (LSM6DS3TR-C: 205 min); very quiet sensors need a longer L to keep the bias insignificant (datasheet grade: Marginal at 109 min with $L = 2048$, Identifiable with $L = 4096$). **Recommendation.** Keep $L = 2048$ and 109 min as the frozen production setting; treat 17 min as a lower bound at p_0 , not a recording target.

1 What the study does

The floe (p_0), layout (Level 2C, reference at $0.9R$), sea state (JONSWAP $H_s = 0.05$ m, $\omega_p = 5$ rad/s) and EMM field are fixed; only the record length T and the Welch segment length L change. For each of the 6 noise cases, 7 durations (10 to 240 min, $\Delta t = 0.1$ s, so $N = 600T$) and 4 segment lengths, the E1 engine gives $\sigma_{\ln \beta}$, $\sigma_{\ln R}$, the Welch-bias significance d_W , $\kappa(F)$ and the class. No new EMM solves are needed. The 109 min point uses the production record ($N = 65536$) and reproduces E1 exactly.

Admissibility. With 50% overlap, a record of N samples gives $K \approx 2N/L - 1$ segments. A combination is admitted only if $K \geq 8$ (pre-registered); below that the spectral matrix is too poorly averaged for the covariance model. 24 of the 28 combinations are admitted:

$K \setminus T$ (min)	10	20	30	60	109	180	240
$L = 512$	22	45	69	139	255	420	561
$L = 1024$	10	22	34	69	127	209	280
$L = 2048$	4	10	16	34	63	104	139
$L = 4096$	1	4	7	16	31	51	69

Grey: $K < 8$, inadmissible, not evaluated.

Why a record can be too long. At fixed L the random error falls with the effective number of averages, while the Welch bias of the transmissibility estimate does not depend on N :

$$\sigma \propto \frac{1}{\sqrt{N_{\text{eff}} - 1}}, \quad \delta\theta_W = \text{const.} \quad \Rightarrow \quad d_W = \sqrt{\delta\theta_W^T F \delta\theta_W} \propto \sqrt{N_{\text{eff}} - 1}.$$

The same bias therefore becomes *more* significant as the record grows. For each L the Identifiable region ($s_{\max} \leq 1\%$, $d_W \leq 0.5$, $\kappa \leq 10^8$) is a window $[T_{\text{lo}}, T_{\text{hi}}]$: too short and σ fails, too long and d_W fails. A short L averages many segments (small σ) but smooths the response over a wide frequency band (large $\delta\theta_W$); a long L resolves the response but needs a long record to reach $K \geq 8$. “Precise” (small σ) is not the same as “identifiable”.

Between grid points. Both scalings were checked on the grid and hold exactly (deviation 0.00% for $s_{\max}\sqrt{N_{\text{eff}} - 1}$ and $d_W/\sqrt{N_{\text{eff}} - 1}$), so the window edges are found on a 0.1 min grid using the exact N_{eff} of each record. The preferred L is the one Identifiable earliest; ties go to the widest window. A secondary measure, $\text{RMSE} = \sqrt{\sigma_{\ln \beta}^2 + \delta \ln \beta_W^2}$, combines both errors.

2 Results

Headline: shortest Identifiable record and the production setting (109 min, $L = 2048$).

Noise case	$\sigma_{\ln \beta}$	d_W	Shortest Identifiable	Preferred L	Production verdict
Datasheet grade	0.12%	0.624	7.7 min	1024	Marginal ($d_W > 0.5$); Identifiable with $L = 4096$
Three IMUs fused	0.28%	0.369	12.0 min	1024	Identifiable; $9.1\times$ minimum
LSM6DSV16X	0.36%	0.309	17.1 min	2048	Identifiable; $6.4\times$ minimum
G3 assumption	0.47%	0.244	26.5 min	1024	Identifiable; $4.1\times$ minimum
ICM-42688-P	0.48%	0.236	28.2 min	1024	Identifiable; $3.9\times$ minimum
LSM6DS3TR-C	1.38%	0.088	204.8 min	1024	Marginal ($\sigma > 1\%$)

$\sigma_{\ln \beta}$ and d_W at the production setting.

Identifiable windows $[T_{\text{lo}}, T_{\text{hi}}]$ in minutes (search to 600 min; $L = 512$: never Identifiable).

Noise case	$L = 1024$	$L = 2048$	$L = 4096$	$\sigma_{\ln \beta} \leq 1\%$ first reached, $L = 2048$
Datasheet grade	[7.7, 12.7]	[15.4, 71.6]	[30.8, 515.4]	15.4*
Three IMUs fused	[12.0, 23.8]	[15.4, 197.9]	[30.8, >600]	15.4*
LSM6DSV16X	[17.1, 29.0]	[17.1, 281.5]	[30.8, >600]	17.1
G3 assumption	[26.5, 40.1]	[27.4, 450.5]	[30.8, >600]	27.4
ICM-42688-P	[28.2, 41.8]	[29.1, 477.8]	[34.2, >600]	29.1
LSM6DS3TR-C	[204.8, 226.9]	[206.6, >600]	[211.7, >600]	206.6

* set by the $K \geq 8$ rule, not by σ (the shortest admissible record already meets 1%). $L = 4096$ cannot start before 30.8 min for the same reason.

RMSE-optimal L ($\ln \beta$) by duration: the optimum moves to longer segments as data accumulates.

T (min)	10	20	30	60	109	180	240
LSM6DSV16X	1024	2048	2048	2048	4096	4096	4096
Datasheet, fused, G3, ICM	1024	1024–2048	2048	2048–4096	4096	4096	4096
LSM6DS3TR-C	512	1024	1024	2048	2048	2048	2048

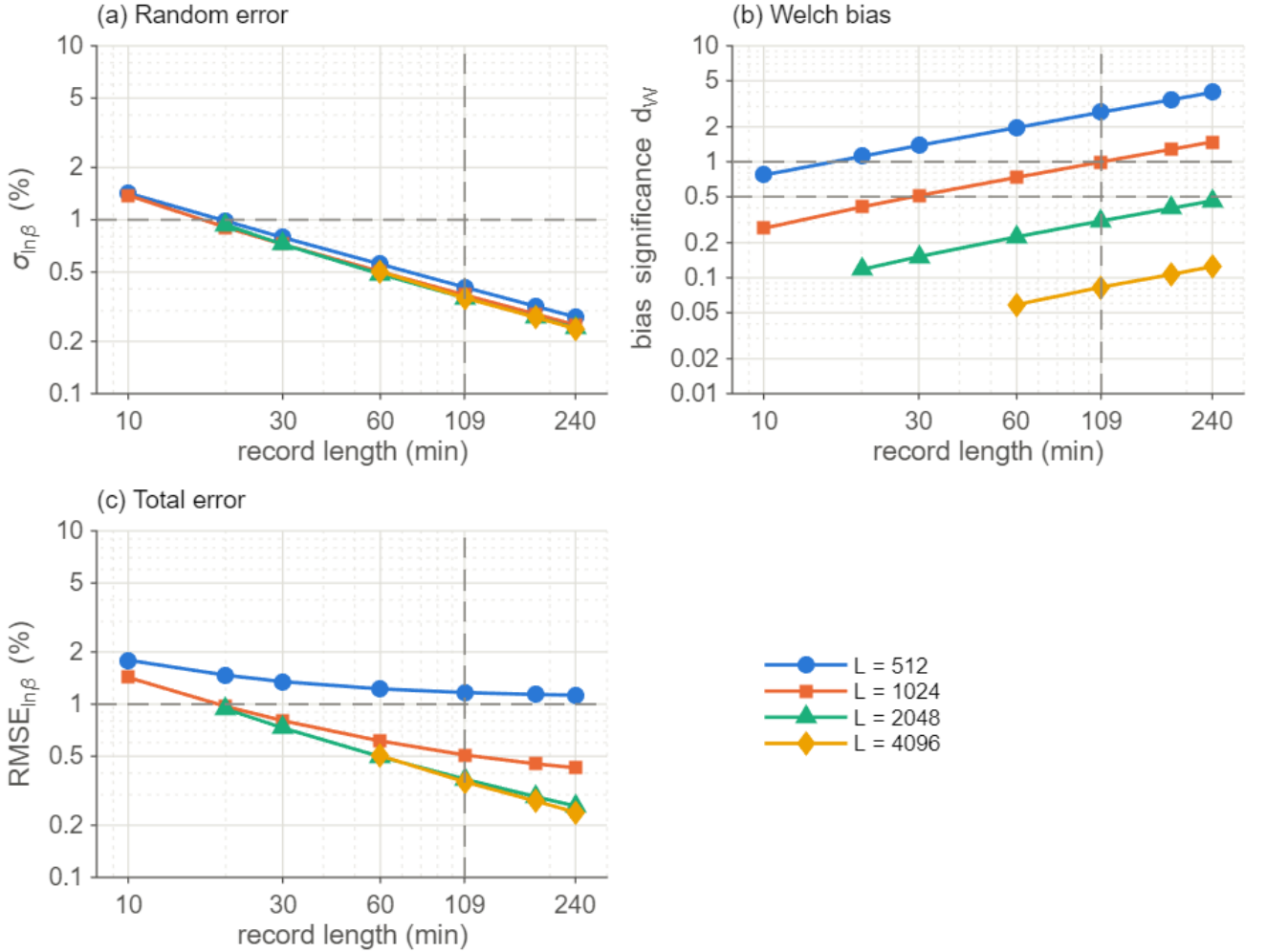


Figure 1: Identifiable record-length windows for each sensor-noise case and Welch segment length L . Each marker denotes the shortest record satisfying the pre-registered Identifiable criteria ($s_{\max} \leq 1\%$, $d_W \leq 0.5$, and $\kappa(F) \leq 10^8$), while the horizontal line shows the duration interval over which those criteria remain satisfied. Arrows indicate cases that remain Identifiable beyond the 600 min search limit. The dashed vertical line marks the 109 min production record. $L = 512$ is omitted because it is never Identifiable for any noise case.

3 Interpretation

1. **$L = 2048$ is justified by robustness, not just precedent.** For the LSM6DSV16X, $L = 1024$ and $L = 2048$ both become Identifiable at 17.1 min, but $L = 2048$ remains so to 281.5 min instead of 29 min. 109 min sits in the middle of the $L = 2048$ window, far from both edges.
2. **$L = 512$ is ruled out.** It is never Identifiable for any noise case: by the time σ is small enough, its window smoothing has already pushed d_W above 0.5 (for the LSM6DSV16X, $d_W = 0.75$ even at 10 min; Figure 1b).

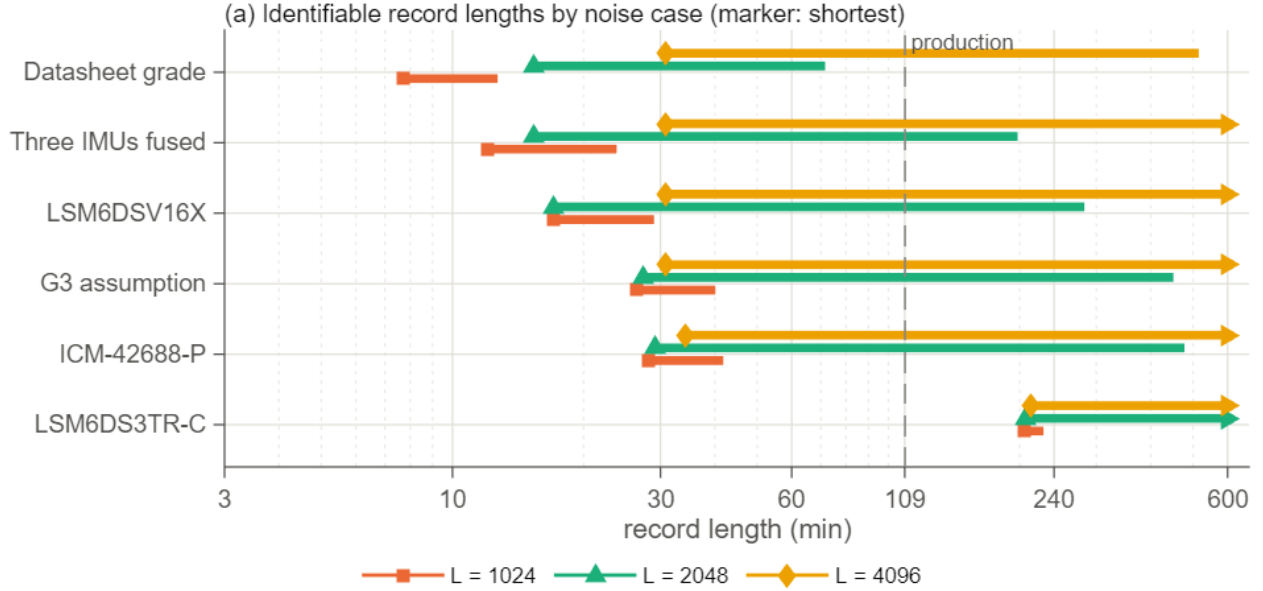


Figure 2: Effect of record duration and Welch segment length on the LSM6DSV16X case. (a) Random uncertainty in the recovered stiffness, $\sigma_{\ln \beta}$, with the 1% Identifiable threshold shown dashed. (b) Welch finite-resolution bias significance d_W , with the $d_W = 0.5$ and $d_W = 1$ thresholds shown dashed. (c) Parameter-space $\text{RMSE}_{\ln \beta} = \sqrt{\sigma_{\ln \beta}^2 + \delta \ln \beta_W^2}$. Only combinations satisfying the $K \geq 8$ Welch-segment requirement are shown. The dashed vertical line marks the 109 min production record.

3. **Two regimes set by the sensor.** A noisy sensor is precision-limited and needs a long record (LSM6DS3TR-C: about 205 min). A very quiet sensor is bias-limited and needs a longer segment: the datasheet-grade case is only Marginal at 109 min with $L = 2048$ ($d_W = 0.62$) and Identifiable with $L = 4096$. Better sensors only pay off if the estimator's resolution keeps up.
4. **$L = 4096$ at 109 min.** It is RMSE-optimal there, but for the LSM6DSV16X its advantage is in bias (d_W about 0.08 instead of 0.31); the RMSE curves for $L = 2048$ and $L = 4096$ nearly coincide (Figure 1c). It is not a reason to change the frozen setting now. $L = 4096$ lies outside the E1 validation gate (512 to 2048), and R2 tests whether the choice of L holds across sea spectra. $L = 4096$ is therefore a candidate for R2 and an M1 spot check.
5. **17.1 min is a lower bound, not a target.** It holds at p_0 , for this sea state and this noise model. E1 is a local Fisher prediction, about 5 to 10% optimistic (G3); other floes (A1), sea states (R2) and real noise (R1) will move it, and a field record is not stationary for hours. The minima marked * sit at the $K \geq 8$ boundary, where the covariance model is least tested (E1 was validated at $K = 63$). In practice, record well beyond the minimum.
6. **The earlier bias rule of thumb is not reliable.** The local rule $d \approx 0.26 \sqrt{N/65536} (2048/L)^2$ overstates the engine's d_W by a factor of 1.01 to 1.81 (G3 noise): the bias does not scale exactly as L^{-2} . The engine values should be used.

Requirement for the experiment and the thesis

At p_0 with the LSM6DSV16X, a record of at least 17 min with $L = 2048$ meets every Identifiable criterion, and $L = 2048$ stays valid up to about 280 min. Keep $L = 2048$ and about 109 min in production: it is well inside the window and leaves margin for the E1 optimism, other floes and non-ideal conditions. Do not use $L = 512$. Revisit $L = 4096$ for long records only after R2 and an M1 check.

Outputs (Results/D2/): D2.20261002.114914.log/.mat/.summary.txt; D2.sweep.csv (every noise case, L , T : K , N_{eff} , σ , d_W , RMSE, κ , bins, class); D2.min.duration.csv (windows, minima, verdicts); D2.rmse.optimal.L.csv; figures/D2_fig1.duration.lsm6dsv16x, figures/D2_fig2.identifiable.window (.pdf/.png).